Survival-Supervised Bayesian Matrix Factorization for Prognostic Molecular Programs in Pancreatic Cancer

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Summary

Aim. Bulk pancreatic tumors mix tumor, stromal and immune signals. We ask whether a Bayesian matrix factorization that lets survival inform its molecular programs can find prognostic programs that an unsupervised factorization misses, and whether its risk score transfers to new cohorts.

Method. The model factorizes expression into patient activities and gene loadings and links survival through a Cox model on the projection of each patient's expression onto the loadings. A survival weight w_S sets how much the Cox partial likelihood is weighted relative to the expression likelihood. The comparator throughout is the same factorization at the same rank fitted without survival, followed by a Cox model on its program scores.

Results. On 273 training patients, five-fold cross-validated concordance was 0.663 for the supervised fit and 0.652 for the two-step fit. Across 5 external series the mean concordance was 0.620 against 0.613, a pooled paired difference of +0.007 (95% donor-bootstrap interval −0.003 to 0.017). On these data survival supplied at most 8×10^{-7} of the loading precision, so the supervised and unsupervised loadings are the same to the reported precision. The difference reflects joint coefficient estimation with a shrinkage prior. A pre-registered October 15 check of an earlier supervised configuration gave a paired difference of −0.003 (95% interval −0.029 to 0.024). Its lower limit fell below the non-inferiority margin. In simulation, supervision matches the two-step fit when a prognostic program is easy to find. Supervision improved recovery near the detection threshold under strong survival. The real-data refit retained the unit weight, and the working fit remains unpruned.

1 Methods

1.1 Notation

Patients are indexed by $i = 1, \dots, n$, genes by $j = 1, \dots, p$ and programs (factors) by $k = 1, \dots, K$. Patient i has observed time t_i and event indicator δ_i (1 for death, 0 for censoring). The training data are an $n \times p$ expression matrix Y. For a random quantity θ under the variational posterior q, $\bar{\theta} = E_q \theta$.

Two patient-level quantities are kept apart. The **activities** $L \in \mathbb{R}^{n \times K}$ are posterior scores in the reconstruction $Y \approx LF^\top$ and exist only for training patients. The **program scores** $Z = Y\bar{F}$ are the product of a patient's expression with the posterior-mean loadings. They enter the Cox predictor and can be computed for any new patient whose expression is measured on the same genes.

1.2 Model

The expression matrix is modeled as

$$y_{ij} = \sum_{k=1}^{K} L_{ik} F_{jk} + e_{ij}, \qquad e_{ij} \sim N(0, \tau_j^{-1}), \tag{1}$$

with independent point-Laplace priors on the entries of each column of L and F, a point mass at zero mixed with a Laplace distribution, whose parameters are estimated by empirical Bayes within each program with the R package ebnm (Willwerscheid et al. 2025). Each gene has its own residual precision τ_j. Survival follows a Cox model (Cox 1972) whose linear predictor is the program-score

projection,

$$\eta_i = \sum_{k=1}^{K} Z_{ik}\beta_k, \qquad Z = Y\bar{F}, \qquad \beta_k \sim N(0, \sigma_k^2), \tag{2}$$

with the prior variance σ_k^2 estimated per program. This is the YFB model, named for its predictor $\eta = (YF)\beta$. Because new patients are scored by the same projection, prediction needs no refitting.

The variational objective is

$$E_q \log p(Y \mid L, F, \tau) \; + \; w_S \, E_q \log \mathrm{PL}(\beta, F) \; - \; \mathrm{KL}(q \,\|\, \mathrm{prior}), \tag{3}$$

where PL is the Cox partial likelihood. The survival weight w_S multiplies the Cox term wherever it appears: in the loading update, in the coefficient update and in the objective. At $w_S = 1$ the objective is the evidence lower bound of the joint model. Larger values of w_S give survival more influence on the loadings. The full derivation, with every update checked term by term against the code, is in `derivations/SSBMF-YFB-model-derivations/SSBMF-YFB-model-derivation-full.pdf` (Appendix A).

Scale. The reconstruction does not fix the scale of a program: multiplying $L_{\cdot k}$ by c and dividing $F_{\cdot k}$ by c leaves $LF^\top$ unchanged. The fit therefore rescales each nonzero program after a sweep so that its posterior mean training projection has standard deviation 1. Each β_k is then a log hazard ratio per standard deviation of the program score, and a single w_S means the same thing for every program.

1.3 Fitting and prediction

Each sweep re-expands the Cox partial likelihood to second order around the current $\bar{\eta}$, with the predictor's variance taken from $q(\beta)$ and $q(F)$. It then updates, one program at a time, the coefficient, the activities and the loadings. The loading update uses the reconstruction residual and the weighted Cox contribution to its precision and linear term. Each sweep ends by updating the noise precisions and rescaling the programs. The fit starts from the singular value decomposition of Y and stops when the relative change in the objective falls below 10^{-5} or after 300 sweeps. The working fit is unpruned at rank 7 (Section 3.7).

New cohorts are standardized gene by gene within cohort and scored on the training genes they measure. The direction of the risk score is the one the fit learns from training survival. There is no post-fit sign correction: a fit oriented the wrong way would have training concordance below 0.5 and is reported as fitted.

1.4 The two-step comparator

The rank-matched comparison uses the same fitter at $w_S = 0$, which removes survival from the loading update and gives the unsupervised empirical-Bayes factorization at the same rank, on the same genes and patients. Because w_S also multiplies the Cox term in the coefficient update, $w_S = 0$ gives $\beta = 0$. The two-step comparator therefore takes the $w_S = 0$ loadings and fits an unpenalized Cox model on the training program scores. It differs from the supervised fit only in whether survival entered the loadings and in how β was estimated.

1.5 Choosing the survival weight

The current real-data configuration uses $w_S = 1$, fixed by five-fold training cross-validation (Section 3.1). Two refinements were then tested in simulation. In the survival-block form, w_S multiplies the Cox term and the Kullback–Leibler term of β together, so β gets an ordinary Bayesian Cox fit while the loadings see survival at weight w_S. The weight is chosen for each dataset from $\{1, 2, 5, 10, 20, 50\}$ by five-fold cross-validated partial log-likelihood, with ties going to the smaller weight. Applying this rule to the real training data (study S-15) is complete (Section 3.6).

1.6 Earlier configuration

The October 16 version of this writeup evaluated a different fitter configuration, in which the loading weight α_F was 0 and survival never entered the loading update. That model is an unsupervised factorization followed by a Cox fit with a shrinkage prior. An expression-only loading update can miss prognostic programs that explain little expression variance. Its external results are kept as a historical reference and are reported here under the corrected preprocessing (Section 3.8).

2 Evaluation

2.1 Data

Table 1: Training and validation cohorts. Patient and death counts are read from the saved fold table and external scoring results. Scores omit training genes absent from the external series. PACA-AU is the Australian pancreatic cancer project of the International Cancer Genome Consortium (ICGC). Its array and RNA-seq series share donors. TCGA-PAAD, The Cancer Genome Atlas pancreatic adenocarcinoma cohort. CPTAC, Clinical Proteomic Tumor Analysis Consortium. RNA-seq, RNA sequencing.

Cohort	Platform	Patients	Deaths	Missing genes	Role
TCGA-PAAD	RNA-seq	144	75	0	Training
CPTAC	RNA-seq	129	64	0	Training
Dijk	RNA-seq	90	81	30	External validation
Moffitt	Microarray	123	83	71	External validation
PACA-AU array	Microarray	63	38	13	External validation
PACA-AU RNA-seq	RNA-seq	52	31	27	External validation
Puleo	Microarray	288	181	82	External validation

The model is trained on 273 patients (139 deaths) from two RNA-seq cohorts, TCGA-PAAD and CPTAC (Table 1). Every RNA-seq cohort is transformed once by $\log_2(x + 1)$. Earlier benchmarks applied this transform twice to some RNA-seq cohorts, an error found in review and corrected on October 30. Genes are screened separately in each training cohort by the combined mean and variance rank of DeSurv (Young et al. 2026), keeping the top 3,000 per cohort. The intersection, called the D4 gene set, has 2,218 genes under the single log. Each cohort is then standardized gene by gene to mean 0 and standard deviation 1.

The 5 external series supply 616 records and 414 recorded deaths. The PACA-AU array and RNA-seq series share 47 donors, and the RNA-seq series has 52 records from 51 donors, so the external collection represents 568 people if the other cohorts are

distinct (counts from the October 16 accuracy review, `paper/writeup/2026-10-16-writeup/_process/Accuracy_Review_Dispositions_10_16_26.md`).

2.2 Uses of the external cohorts

Earlier comparisons across the external series informed several choices inherited by the current model. Table 2 records the development and evaluation uses documented in the audit.

Table 2: Every use of the five external series in a model-development choice, from the project decision log. "Chosen outright" means external concordance was the deciding criterion. "Informed" means it was one of several pieces of evidence. C is Harrell's concordance index. LASSO is the least absolute shrinkage and selection operator.

Choice	Date	How the external series entered	In the current model	Record
Historical coefficient prior	April 24	External concordance and simulation favored point-Normal over point-Laplace	No	DECISIONS.md, 2026-04-24
Initialization constraints	May 6	External C informed discarding the nonnegative and Cox-aligned starts alongside training ELBO and factor counts	No	DECISIONS.md, 2026-05-06
July methodological changes	July 12–15	An external net-benefit check followed correction of a mismatch with the training preprocessing recipe	Historical	DECISIONS.md, 2026-07-12 and 2026-07-15
Within-cohort standardization, no rank transform	May 25	Chosen outright: highest mean external C	Yes	DECISIONS.md, 2026-05-25
D4 gene screen (per-cohort combined mean and variance rank)	May 27	Chosen outright: adopted because external C rose by more than a set threshold	Yes	DECISIONS.md, 2026-05-27
Pooled fit without cohort indicators	May 22–27	Informed: external C compared alongside a training diagnosis	Yes	DECISIONS.md, 2026-05-22 to 2026-05-27
Starting rank	July to August	Training CV proposed the rank. External performance informed retaining it over smaller starts and warm-start or optimization alternatives	Yes	DECISIONS.md, 2026-08-19 and 2026-08-27
Loading weight $\alpha_F = 0$ and frozen-loading start N_{frozen}	August 20	The defaults predate external use. An external grid review retained both defaults	No: the current fitter uses w_S	DECISIONS.md, 2026-08-20
Historical two-step comparator	July to August	External C compared rank and penalization configurations. Training CV chose the LASSO penalty	No: replaced by the $w_S = 0$ two-step fit	DECISIONS.md, 2026-08-20
October 15 supervised configuration	October 15	Chosen in simulation, then scored once against a registered criterion (Section 3.2)	Superseded	DECISIONS.md, 2026-10-15
Single-log correction	October 26	External sensitivity quantified the effect of correcting the duplicate log. The correction followed the training recipe	Yes (single log)	Single-log check
Current supervised fit	November 5	S-11 fixed the weight, feature-set arm and screening arm by training CV before S-12 scoring (Section 3.1)	Yes	S-12 findings

The gene screen, the standardization and the rank of the current model therefore all descend from choices made with external concordance. Every external series informed at least one earlier development choice. The training cross-validation of Section 3.1 fixed only the survival weight, the gene-set size and the screening arm, within this inherited setup. Holding back one cohort from every remaining choice is listed under continuing work. The dated external-cohort audit (`docs/findings/2026-11-06-external-cohort-uses.md`) documents earlier decisions and distinguishes them from subsequent evaluation.

2.3 Simulation designs

All three simulation studies plant a single prognostic program among unsupervised structure and ask whether the fit recovers it. Recovery requires absolute loading correlation of at least 0.7, an active coefficient and the correct coefficient sign after loading alignment. A **false-active** program is one whose coefficient lies more than 1.96 posterior standard deviations from zero but which does not match the planted program.

- **Survival weight at 2,000 genes (S-9).** Data with 50 or 2,000 genes and a planted program explaining 0.25%, 1.00%, 5.00% of expression variance. We fitted 10 datasets per cell at $w_S \in \{0, 1, 10, 100, 1{,}000, 10{,}000\}$, at $w_S = p/50$ and at a weight calibrated to give survival 20% of the loading precision. The generator uses strong planted survival effects.
- **Weight chosen by cross-validation (S-13).** The larger gene-count cells of S-9 plus a cell with weaker survival. The weight was chosen per dataset by the rule of Section 1.5. Two arms differ in the coefficient posterior: mean-field (A1) or a joint Gaussian posterior for β (A2).
- **Programs near the detection threshold (S-14).** Planted programs explained 0.05%, 0.10%, 0.20% of expression variance, with 10 datasets per cell. The mean oracle concordance ranges were 0.94 to 0.96 for strong survival and 0.73 to 0.74 for realistic survival.

S-13 and S-14 compare the supervised and matched two-step fits on the same datasets. The S-9 registration used an earlier uncentered unsupervised reference. The matched two-step comparison on the S-9 cells comes from S-13.

2.4 Performance metrics and uncertainty

Discrimination is Harrell's concordance index C (Harrell et al. 1982), with the risk direction fixed from training. Paired differences between two models use percentile bootstrap intervals (Efron and Tibshirani 1993) from resampling patients within each cohort, with both models scored on the same resamples. The pooled difference is the equal-weight mean of the within-series differences. Calibration is the slope from a Cox regression of held-out survival on the fitted score. A slope above one indicates fitted risk differences that are too small. A slope below one indicates differences that are too large. In simulation the slopes use exact event times, because the default tie handling of `survival::coxph()` merges the extreme simulated times and flattens every slope.

The original external bootstrap resamples records within series. The donor sensitivity resamples each donor together with every observed record, so the 47 donors shared by the two PACA-AU series move together. Both pooled intervals use the same equal-weight mean of the within-series differences.

3 Results

3.1 Supervised fit against the rank-7 two-step fit

Table 3: Five-fold training cross-validation on the 2,218 D4 genes (single log). Both fits have rank 7 and use the same folds. The two-step fit takes the loadings of the w_S = 0 fit and fits a Cox model on the training program scores. Survival-active: coefficient at least 1.96 posterior standard deviations from zero in the fit to all training patients (Wald test for the two-step fit). Convergence refers to the factorization. C is Harrell's concordance index. SE is standard error across folds.

Fit	Mean fold C (SE)	Pooled out-of-fold C	CV partial log-likelihood	Folds converged	Survival-active
Supervised, w_S = 1	0.663 (0.024)	0.663	−775.2	3 of 5	3
Two-step (w_S = 0 loadings, then Cox)	0.652 (0.026)	0.647	−777.1	3 of 5	4

On the training patients, the supervised fit had mean fold concordance 0.663 against 0.652 for the two-step fit (Table 3). The paired fold difference was +0.011 (SE 0.005). In 2 of the five folds both fits reached the 300-sweep cap with the objective still changing slowly. Their coefficients were small, and no coefficient or noise precision was growing, so these folds were moving slowly near their optimum rather than diverging.

This configuration ($w_S = 1$, the D4 genes, no survival screen) was fixed from these training results before any external cohort was scored. It was then scored once on the five external series (Table 4).

Table 4: External concordance of the fixed supervised fit and the two-step fit. Each series was scored once, after the configuration was fixed by training cross-validation. C intervals are 95% percentile bootstrap intervals over patients. The difference is supervised minus two-step on the same resamples. The pooled comparison reports intervals from resampling records and resampling donors, plus a sensitivity excluding PACA-AU RNA-seq. The calibration slope is the coefficient of held-out survival on the fitted risk score; 1 is calibrated.

Series	C, supervised (95% CI)	C, two-step	Difference (95% CI)	Slope, supervised	Slope, two-step
Dijk	0.584 (0.513 to 0.658)	0.567	+0.018 (0.000 to 0.035)	0.55	0.35
Moffitt	0.537 (0.459 to 0.611)	0.531	+0.007 (−0.014 to 0.026)	0.36	0.28
PACA-AU array	0.656 (0.562 to 0.751)	0.656	0.000 (−0.026 to 0.025)	0.78	0.73
PACA-AU RNA-seq	0.681 (0.579 to 0.780)	0.671	+0.010 (−0.019 to 0.038)	1.57	1.26
Puleo	0.644 (0.599 to 0.686)	0.643	+0.001 (−0.007 to 0.010)	0.99	0.80
Mean of five series	0.620	0.613			
Pooled, record bootstrap			+0.007 (−0.002 to 0.017)		
Pooled, donor bootstrap			+0.007 (−0.003 to 0.017)		
Pooled, without PACA-AU RNA-seq			+0.006 (−0.003 to 0.016)		

The supervised fit scored equal to or higher than the two-step fit in every series. Its mean external concordance was 0.620 and the two-step fit's 0.613. The pooled paired difference was +0.007 (95% interval −0.002 to 0.017), and +0.006 (−0.003 to 0.016) without the PACA-AU RNA-seq series. The donor bootstrap interval is −0.003 to 0.017 for the full panel. All pooled intervals include zero. The difference is about the size of the training cross-validation difference. Dijk and Moffitt were the weakest series for both fits, as they were for every earlier model.

The supervised slope was closer to one in 4 of 5 series. In PACA-AU RNA-seq it was 1.57, compared with 1.26 for two-step. In Puleo it was 0.99.

As a summary on the scale DeSurv reports (Young et al. 2026), the hazard ratio per standard deviation of the risk score, from a Cox model stratified by series over all 616 records, was 1.37 (95% CI 1.24 to 1.51) for the supervised fit and 1.35 (1.22 to 1.49) for the two-step fit.

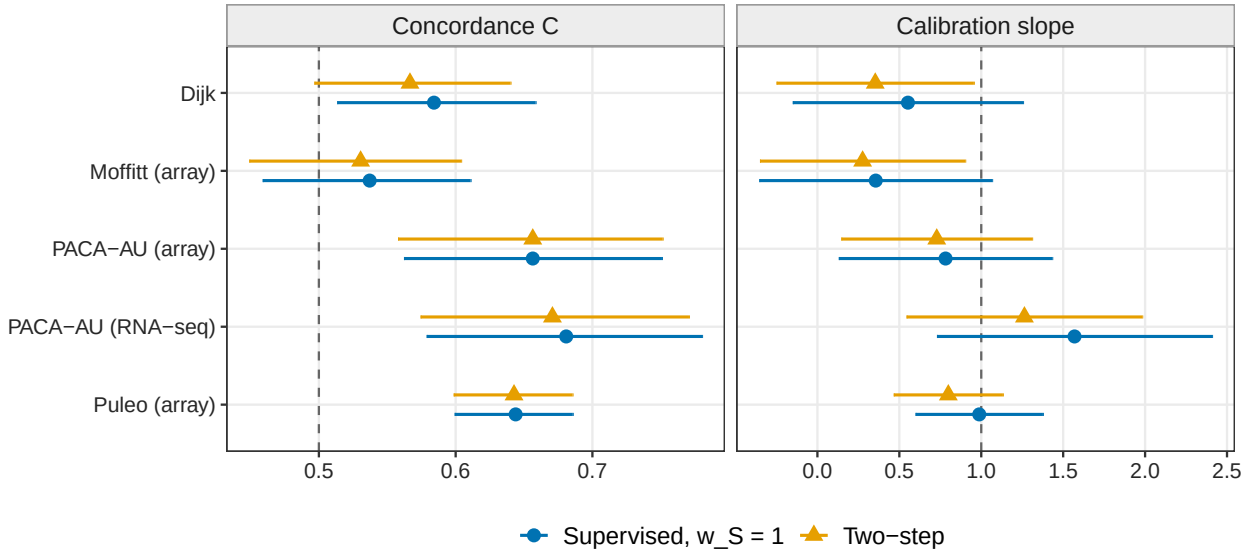


Figure 1: External concordance with bootstrap intervals and calibration slopes with Wald intervals for the two fixed fits.

3.2 The pre-registered October 15 supervision check

On October 15 an earlier supervised configuration was evaluated against a criterion fixed before it ran. That configuration used the same predictor with $w_S = 1$, a supervised start and a shared point-Laplace prior on β, on the 2,064 genes of the historical preprocessing. In simulation it recovered a planted low-variance prognostic program in 17 of 20 datasets at the lower noise level and 19 of 20 at the higher one. The registered real-data estimand was the mean over the 5 external series of the per-series difference in C from the frozen $\alpha_F = 0$ benchmark, with a cohort-stratified paired bootstrap. The criterion was a lower 95% limit above −0.02.

The estimate was −0.003 (95% interval −0.029 to 0.024). The lower limit missed the margin, so **the criterion was not met**. The interval included both a loss larger than the margin and a gain in concordance. Mean external C was 0.624 for the supervised fit and 0.627 for the benchmark. After the KL normalizer was corrected later the same day, the refit gave −0.006 (−0.032 to 0.021),

which also misses the margin. The registered values are read from the result file as committed at the time (commit `65cf90a`), and the rerun values from the current file. Work continued with the configuration recorded as not meeting the criterion (DECISIONS.md, 2026-10-15).

3.3 What survival does to the real-data loadings

On the training data at $w_S = 1$, survival supplies between 2×10^{-10} and 8×10^{-7} of the loading precision (median over programs, across the gene-set sizes below). The simulation cell medians at the same weight ranged from 0.01 to 0.20 at the larger gene count. Two things keep it small on real data. The fitted coefficients are small (largest $|\bar{\beta}_k|$ 0.41 per standard deviation of the program score), and the share scales with $E[\beta_k^2]$. The real programs also explain much more expression variance than the planted programs in simulation, so their expression precision is large. The loading correlations between the fits round to 1.000 for every matched pair. The concordance difference in Section 3.1 reflects the joint coefficient update with its shrinkage prior.

Table 5: Training cross-validation by input gene-set size. Sets are nested and outcome-free, ranked by the per-cohort mean and variance rank. D4 is the 2,218-gene default. Match: absolute loading correlations of the two survival-active programs of the all-genes supervised fit with their best matches at each size.

Genes	Supervised C (SE)	Two-step C (SE)	Folds converged, supervised	Match
2,218	0.663 (0.024)	0.652 (0.026)	3	0.78 0.89
3,000	0.663 (0.026)	0.653 (0.027)	3	0.72 0.80
5,000	0.657 (0.030)	0.652 (0.031)	4	0.69 0.77
8,001	0.647 (0.030)	0.650 (0.033)	5	0.74 0.97
12,001	0.643 (0.037)	0.653 (0.037)	5	0.96 0.80
17,900	0.648 (0.038)	0.652 (0.037)	5	1.00 1.00

Adding genes beyond the D4 set did not help (Table 5). Supervised concordance ranged from 0.643 to 0.663 across sizes, and every size was within one standard error of the others. The registered rule, which picks the smallest set within one standard error of the all-genes concordance whose survival-active programs match the all-genes fit at $|r| \geq 0.7$, chose D4 for both fits. The supervised advantage over the two-step fit shrank as genes were added and reversed at larger gene-set sizes.

Screening genes on survival inside each training fold, by a cohort-stratified univariable Cox test, gave no consistent gain over an outcome-free screen of the same size: the paired differences were 0.001 to 0.021 across sizes and fits, each within about one paired standard error, and no screened set beat the supervised fit on all D4 genes. With survival permuted in simulated data, the $w_S = 1$ fit had 0.1 to 0.5 false-active programs per dataset on average, below the registered limit of one.

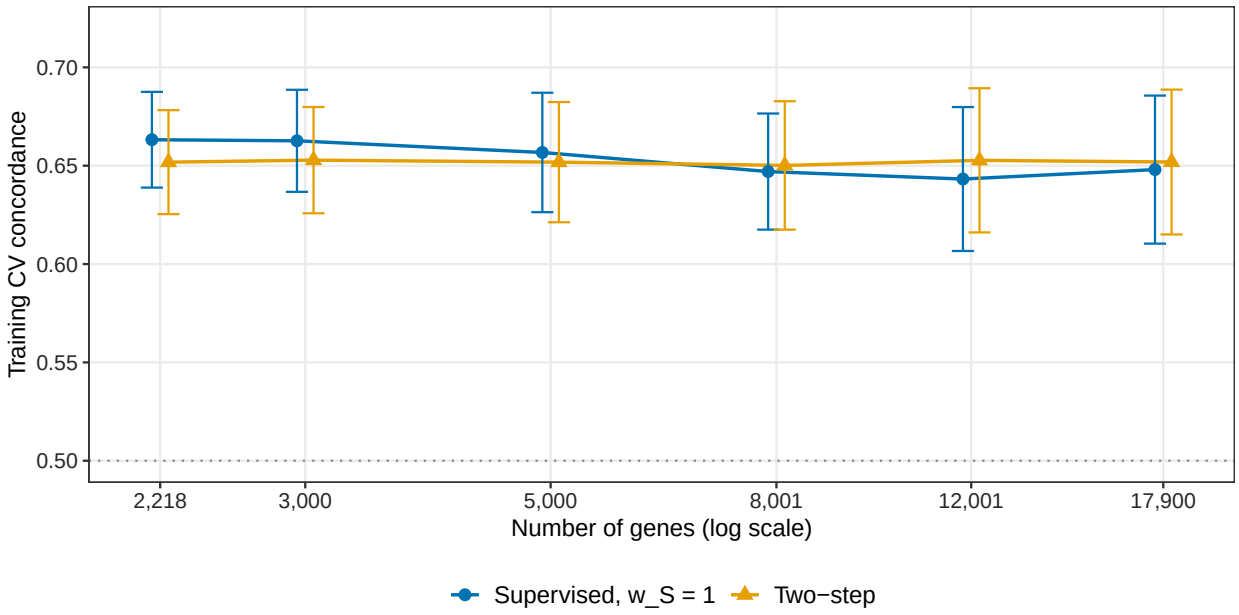


Figure 2: Cross-validated training concordance by gene-set size. Points give fold means with one standard error.

3.4 Simulation: supervision strength and the weight choice

At $w_S = 1$ with 2,000 genes the fit recovered the planted program in all 10 datasets per variance cell, with mean false-active counts 0.9, 0.1, 0.0, and every fit converged. No fit at a weight of 40 or greater converged at the larger gene count. Training CV selected the unit weight in 55 of 60 datasets. The coefficients at $w_S = 1$ were attenuated: median validation calibration slopes were 1.45, 2.16, 3.23 across the three cells, where one denotes calibration. Mean validation concordance ranged from 0.944 to 0.986.

Table 6: Weight chosen by five-fold cross-validated partial log-likelihood (S-13), against the two-step fit on the same datasets. Cells v0.0025, v0.01 and v0.05 are the 2,000-gene S-9 cells by planted variance share. The weak cell uses weaker survival. Ten datasets per cell. Paired entries show supervised / two-step. FA is the mean false-active count. C is mean validation concordance. Slope is the median calibration slope with exact event times.

Arm	Cell	Weights (count)	Recovered	FA	C	Slope
A1	v0.0025	1:1 2:9	10 / 10	0.9 / 1.7	0.946 / 0.939	1.24 / 1.16
A2	v0.0025	2:10	10 / 10	0.6 / 1.7	0.946 / 0.939	1.22 / 1.16
A1	v0.01	5:10	10 / 10	0.0 / 0.9	0.972 / 0.972	1.27 / 1.10
A2	v0.01	5:10	10 / 10	0.0 / 0.9	0.972 / 0.972	1.28 / 1.10
A1	v0.05	10:10	10 / 10	0.0 / 0.2	0.986 / 0.986	1.45 / 0.96
A2	v0.05	10:10	10 / 10	0.0 / 0.2	0.986 / 0.986	1.45 / 0.96
A1	weak	1:8 2:2	10 / 10	1.2 / 0.5	0.671 / 0.668	1.05 / 1.04
A2	weak	1:10	10 / 10	0.9 / 0.5	0.672 / 0.668	1.05 / 1.04

In S-13 the weights selected by partial log-likelihood increased with the planted signal (Table 6). Both supervised arms and the matched two-step fit recovered the planted program in every dataset.

Mean validation concordance differed by at most 0.007. Each arm met the calibration band in 1 of 3 variance cells. The mean-field arm's largest mean null false-active count at its modal weight was 1.1. Neither arm passed all criteria. The mean concordance of the joint and mean-field coefficient variants differed by at most 0.001.

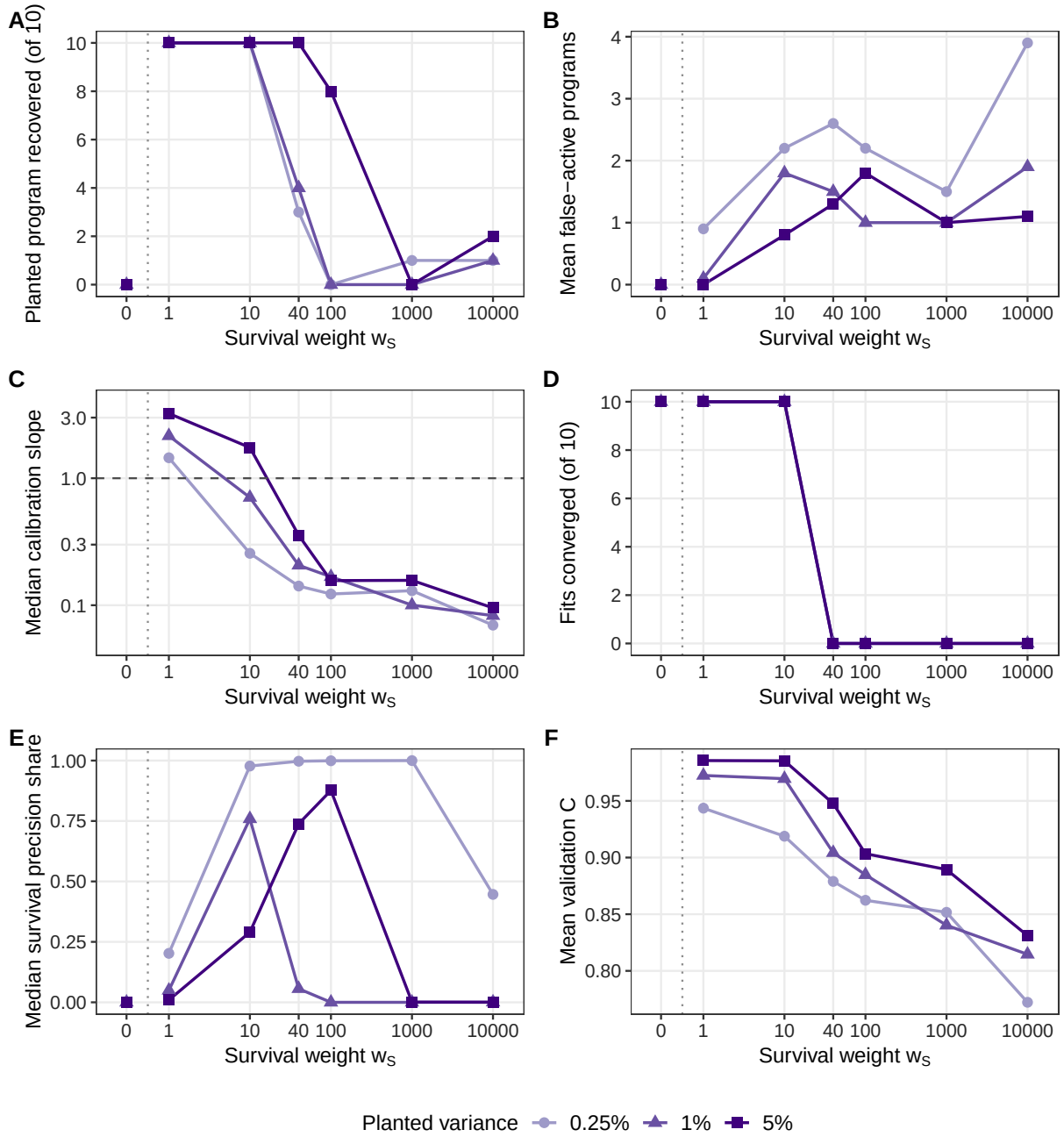


Figure 3: S-9 at the larger gene count. The panels show recovery, false activity, exact-time calibration, convergence, loading precision share and validation concordance.

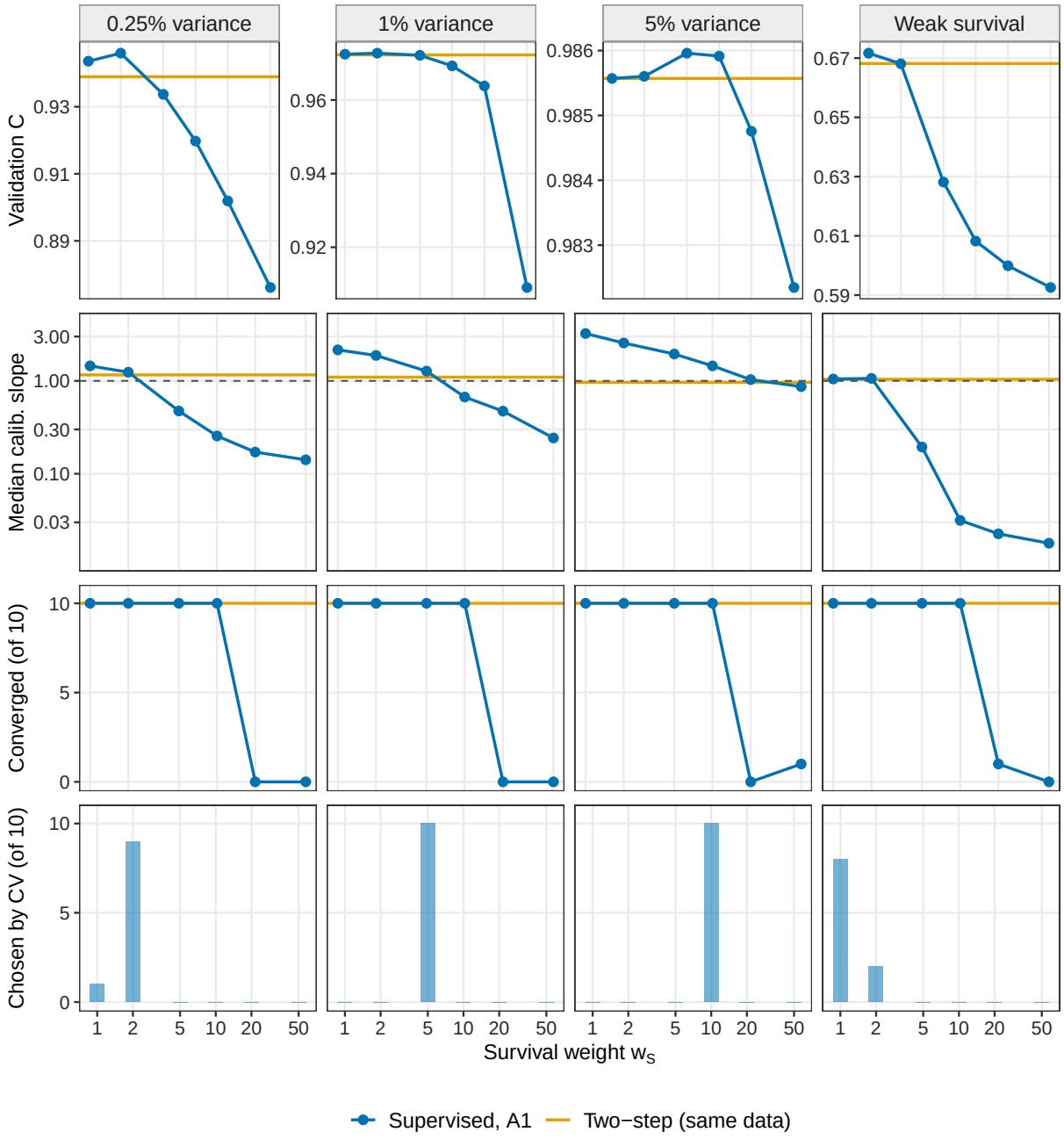


Figure 4: S-13 validation concordance, calibration and convergence by weight, with the matched two-step reference and the CV selection frequencies. The concordance range varies by cell.

3.5 Simulation: programs near the detection threshold

Table 7: Programs near the detection threshold (S-14). All fits converged. Planted share is the planted program's share of expression variance. Entries show supervised / two-step for recovery, false activity (FA) and C. Difference is the mean paired validation C of the supervised fit at w_S = 1 minus the two-step fit, with the number of datasets in which the supervised fit was higher. Oracle C is the validation concordance of the true predictor.

Survival	Planted share	Recovered	FA	C	Difference (wins)	Oracle C
realistic	0.05%	0 / 0	1.9 / 1.6	0.630 / 0.592	+0.038 (9)	0.740
realistic	0.10%	9 / 8	1.0 / 1.1	0.712 / 0.705	+0.008 (8)	0.736
realistic	0.20%	10 / 10	0.5 / 0.8	0.728 / 0.727	+0.002 (8)	0.735
strong	0.05%	5 / 0	2.5 / 2.8	0.800 / 0.644	+0.156 (10)	0.936
strong	0.10%	10 / 8	1.6 / 2.4	0.901 / 0.861	+0.041 (10)	0.948
strong	0.20%	10 / 10	1.0 / 1.8	0.936 / 0.928	+0.008 (9)	0.958

In the S-9 and S-13 cells the two-step fit already finds the planted program, so those studies cannot show what supervision adds. S-14 planted weaker programs (Table 7). With strong survival the supervised fit had the higher validation concordance in 29 of 30 datasets, and the gain grew as the program weakened. At the smallest planted share, 0.05%, the supervised fit recovered the program in 5 of 10 datasets, where the two-step fit recovered none, and validation concordance rose by +0.156. With realistic survival the two fits were close at the larger shares, and neither recovered the program at the smallest share. The registered recovery criterion still failed in every strong cell, because it also required a mean of at most one false-active program per cell. The two-step fit had as many or more false-active programs in the same cells.

The post hoc analysis measures the variance of the sum of the nonmatched score contributions relative to the variance of the full score. It includes nonmatched programs below the survival-activity threshold. It also measures full-minus-matched concordance on the same datasets. In S-13 the largest cell median variance ratio was 0.8%, and the largest absolute cell mean change in concordance was 0.002. In the larger-share S-14 cells the corresponding ratios reached 1.3% for supervision and 2.9% for two-step. Across both arms the largest absolute mean change in C in those cells was 0.008. At the smallest planted share the median ratios ranged from 8.7% to 68.8%. These score summaries leave the registered false-active verdicts unchanged.

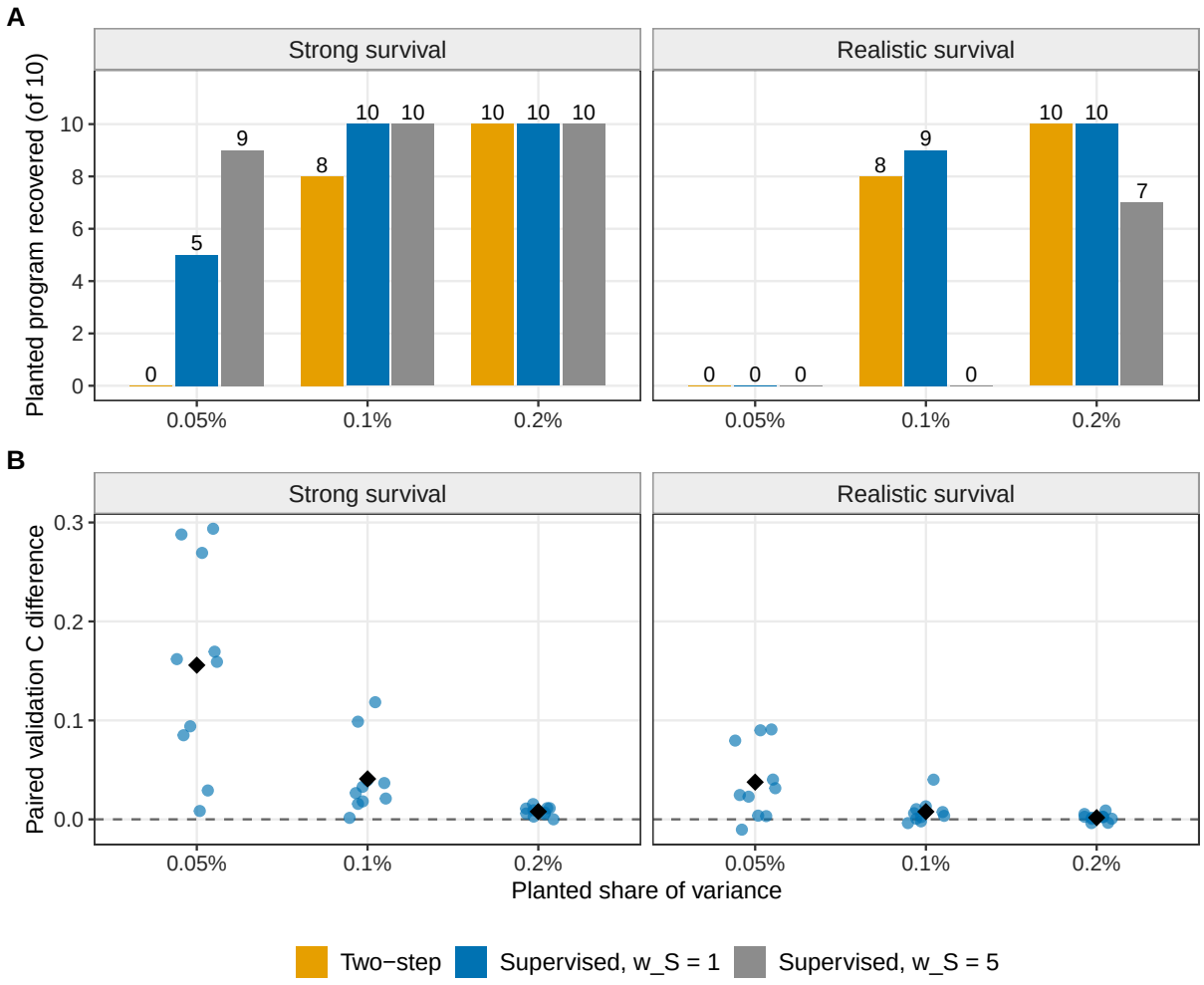


Figure 5: S-14 recovery by arm and paired validation concordance differences for the unit-weight fit versus two-step. Dots in the difference panels represent datasets. Diamonds give cell means.

3.6 Real-data refit at the cross-validated weight

The largest cross-validated partial log-likelihood on the training folds occurred at $w_S = 1$. Across the tested grid from 1 to 50, the criterion varied by 0.26. The nested folds selected weights 5, 2, 50, 1, 10. Nested cross-validation gave mean C 0.656 (SE 0.027), compared with 0.663 for the existing fit. At the largest tested weight the median survival share for active loadings was 7.3×10^{-5}. The registered rule retained the existing fit. The external evaluation is the fixed S-12 comparison in Section 3.1.

Table 8: Training cross-validation of the survival-block weight. The precision share is the median among active programs. The nested evaluation uses weights chosen within each outer training set.

Weight	CV partial log-likelihood	Mean C (SE)	Folds converged	Survival precision share
1	−775.168	0.663 (0.024)	3/5	1.17e-06
2	−775.179	0.663 (0.025)	3/5	2.34e-06

Table 8: Training cross-validation of the survival-block weight. The precision share is the median among active programs. The nested evaluation uses weights chosen within each outer training set.

Weight	CV partial log-likelihood	Mean C (SE)	Folds converged	Survival precision share
5	−775.213	0.661 (0.025)	4/5	5.91e-06
10	−775.244	0.661 (0.025)	4/5	1.20e-05
20	−775.380	0.659 (0.026)	4/5	2.49e-05
50	−775.429	0.653 (0.028)	4/5	7.26e-05

3.7 Rank and ELBO pruning

KL-2 repeated the pruning comparison after correcting the coefficient update. For a flat coefficient likelihood the update returns the prior moments and an exact zero coefficient KL. Ordinary and protected pruning both failed the registered recovery and rank criteria. The working fit retains all 7 starting columns.

Table 9: KL-2 pruning comparison after the flat coefficient-likelihood correction. Convergence refers to the returned fit. The saved summary gives the count of planted-program recoveries and the median retained rank.

Arm	Noise	Recovered	Mean false-active	Converged	Median rank
ordinary	0.25	2/20	1.10	20/20	2
protected	0.25	2/20	1.10	20/20	2
reference	0.25	1/20	1.15	20/20	7
unpruned	0.25	17/20	1.05	20/20	7
ordinary	1.00	2/20	0.20	20/20	1
protected	1.00	2/20	0.20	20/20	1
reference	1.00	1/20	0.40	20/20	7
unpruned	1.00	19/20	0.35	20/20	7

3.8 The $_{\mathbf{F}} = 0$ benchmark under the single log

Table 10: External concordance of the earlier alpha_F = 0 benchmark (rank 7, survival absent from the loading update), as frozen under the double log and refitted with every cohort logged once. The difference interval is a 95% percentile bootstrap over patients within each series. Risk-score correlation compares the two fits' external scores.

Series	n	Frozen (double log)	Single log	Difference (95% CI)	Risk-score correlation
Dijk	90	0.634	0.630	−0.004 (−0.012 to 0.002)	0.991
Moffitt	123	0.549	0.540	−0.009 (−0.019 to 0.001)	0.997
PACA-AU array	63	0.648	0.645	−0.003 (−0.018 to 0.013)	0.998
PACA-AU RNA-seq	52	0.657	0.664	+0.007 (−0.004 to 0.022)	0.997
Puleo	288	0.645	0.645	+0.001 (−0.003 to 0.004)	0.998

The earlier benchmark's external numbers were computed under the double log. Refitted with every cohort logged once and nothing else changed, its mean external concordance was 0.625 against 0.627 frozen, every per-series interval included zero, and the risk scores correlated at 0.99 or more

(Table 10). The single-log number is the one to compare with. It is not a paired comparison with the supervised fit of Section 3.1, which uses a different fitter, input scaling and gene set. After the KL fix, all 5 single-log folds converged. Training cross-validated C was 0.640 with the single log and 0.608 with the double log.

The October 16 writeup led with a comparison against a 40-factor unsupervised factorization followed by LASSO Cox (0.627 against 0.601). That comparator was run only under the double log, and its factor cap was not matched to the model, so it is not used here. Against the rank-matched two-step model the earlier benchmark's advantage was 0.011 (95% interval −0.013 to 0.033), the same small size as the supervised fit's advantage in Section 3.1.

4 Programs of the supervised fit

The joint fit retained 7 programs. Of those, 3 were survival-active (Table 11). Program 2 had a protective coefficient at its positive pole, where the leading genes included classical epithelial and exocrine markers. Its negative pole included hypoxia and glycolysis genes. Program 3 had an adverse coefficient at its ribosomal and proliferative pole. Program 5 also had an adverse coefficient, with enrichment of mitotic and stromal-immune signatures.

Table 11: Programs of the supervised training fit. Coefficients are log hazard ratios per SD of the program score, with posterior SDs in parentheses. The positive and negative poles list the genes with the largest signed loadings in each direction.

Program	Coefficient (SD)	Active	Positive pole	Negative pole
1	0.000 (0.000)	no	TCF4, FBN1, BNC2, GAS7	RAB25, PKP3, GALE, RASSF7
2	−0.405 (0.084)	yes	ATP2A3, ACACB, METTL7A, ADH1B	LOXL2, MMP14, P4HA1, LDHA
3	0.218 (0.073)	yes	RPS7, RPL6, NCAPD2, EEF1D	KIAA1211, SYTL2, CTSE, TSPAN8
4	0.154 (0.082)	no	CTRB2, CTRB1, CPB1, REG1B	BANK1, STK17B, EMB, MS4A1
5	0.230 (0.082)	yes	UHMK1, ARHGAP5, TAOK1, RAD21	CHPF, RCN3, MXRA8, LOXL1
6	0.000 (0.000)	no	IFI30, LILRB4, FCER1G, CD74	RAP1GAP2, IGF1R, ARHGAP32, PBX1
7	−0.048 (0.055)	no	ANO1, BCL9L, PPL, VSTM2L	PRDX4, PPIB, NUCB2, P4HB

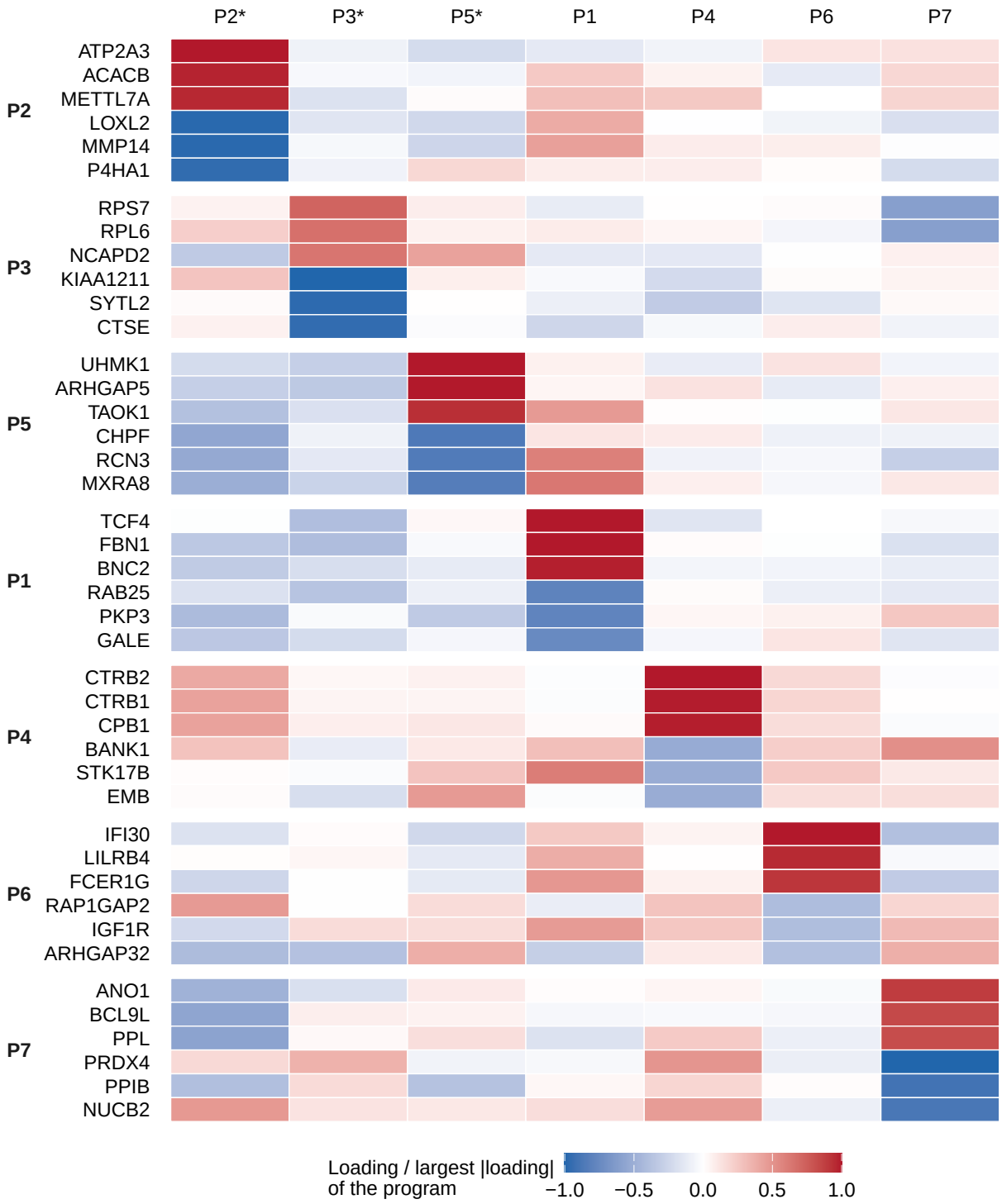


Figure 6: Scaled loadings of selected genes across the fitted programs. Asterisks indicate survival-active programs.

4.1 Program interpretation and correspondence

Enrichment was calculated on signed loadings among the modeled genes. The leading sets are in Table 12. Program 3's influenza-labeled sets share leading ribosomal genes in the saved enrichment results.

Table 12: Leading enrichment at each pole of the survival-active programs. NES denotes normalized enrichment score. The p-values are adjusted within program.

Program	Collection	Gene set	NES	Adjusted p
2	PDAC	DeSurv D1 ClassicalTumor	2.83	9.91e-21
2	Hallmark	EPITHELIAL MESENCHYMAL TRANSITION	−2.82	1.54e-17
3	Reactome	INFLUENZA INFECTION	4.64	4.35e-44
3	PDAC	Moffitt Classical	−2.28	3.22e-10
5	PDAC	DeSurv D2 StromalImmune	2.85	3.80e-30
5	Hallmark	EPITHELIAL MESENCHYMAL TRANSITION	−2.06	3.47e-06

5 Reactome p-values were unestimable and are retained as missing in the source CSV. Spearman correlations with published signatures were computed within series using the available standardized expression genes. In the pooled training data, Programs 2 and 3 were inversely associated with the Moffitt basal-like and classical signatures, respectively. The corresponding external ranges are in Table 13. Correlations between Program 5 and the stromal-immune signature were weaker. These descriptive associations characterize fixed programs.

Table 13: Spearman correlations with selected subtype and stromal signatures. The external interval is the range of point estimates across series. PurIST uses the saved basal-like probabilities.

Program	Signature	Training	External	External PurIST
2	Moffitt basal-like	−0.71	−0.78 to −0.71	−0.64 to −0.45
3	Moffitt classical	−0.76	−0.84 to −0.56	0.30 to 0.79
5	DeSurv stromal-immune	0.21	−0.03 to 0.23	−0.25 to −0.13

The supervised and unsupervised loading correlations on the shared D4 input were close to one (Figure 7). The historical factorization used different preprocessing. The strongest historical match for each active program is reported in Table 14. Overlaps are calculated after aligning the loading signs.

Table 14: Strongest historical loading match for each survival-active program. The Pearson correlation is signed. The gene overlap uses the top absolute loadings after aligning signs.

Program	Historical	Correlation	Top-gene overlap
2	7	−0.623	9
3	3	−0.649	3
5	5	−0.180	0

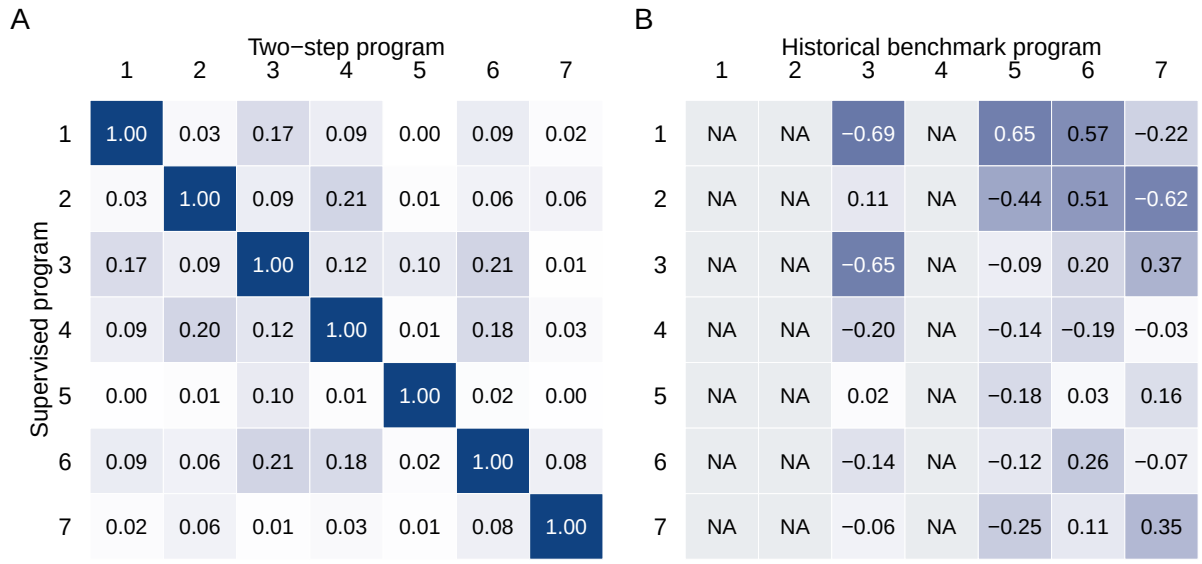


Figure 7: Absolute loading correlations with the matched unsupervised fit and signed correlations with the historical fit on shared genes.

4.2 External program associations and risk groups

Separate external Cox regressions give descriptive associations per series SD of each active program score. The numbers of estimates in the training direction, in program order, were 5, 4, 2 of 5. These estimates are marginal. The training coefficients condition on the other program scores.

Table 15: External program associations from separate Cox regressions. Entries are hazard ratios and Wald intervals per SD of the program score within series.

Series	Program 2	Program 3	Program 5
Dijk	0.64 (0.49 to 0.83)	0.98 (0.79 to 1.21)	0.77 (0.62 to 0.96)
Moffitt	0.90 (0.73 to 1.12)	1.12 (0.90 to 1.40)	0.87 (0.69 to 1.09)
PACA-AU array	0.78 (0.58 to 1.06)	1.28 (0.98 to 1.68)	1.08 (0.79 to 1.47)
PACA-AU RNA-seq	0.71 (0.51 to 0.99)	1.84 (1.31 to 2.59)	1.21 (0.82 to 1.79)
Puleo	0.72 (0.61 to 0.85)	1.37 (1.18 to 1.59)	0.93 (0.81 to 1.07)

Kaplan–Meier curves use within-series tertiles of the frozen risk score or fixed program scores. Program scores are oriented using the training coefficients. The tertiles describe relative ranking within each series. They use a different set of cutpoints in each series.

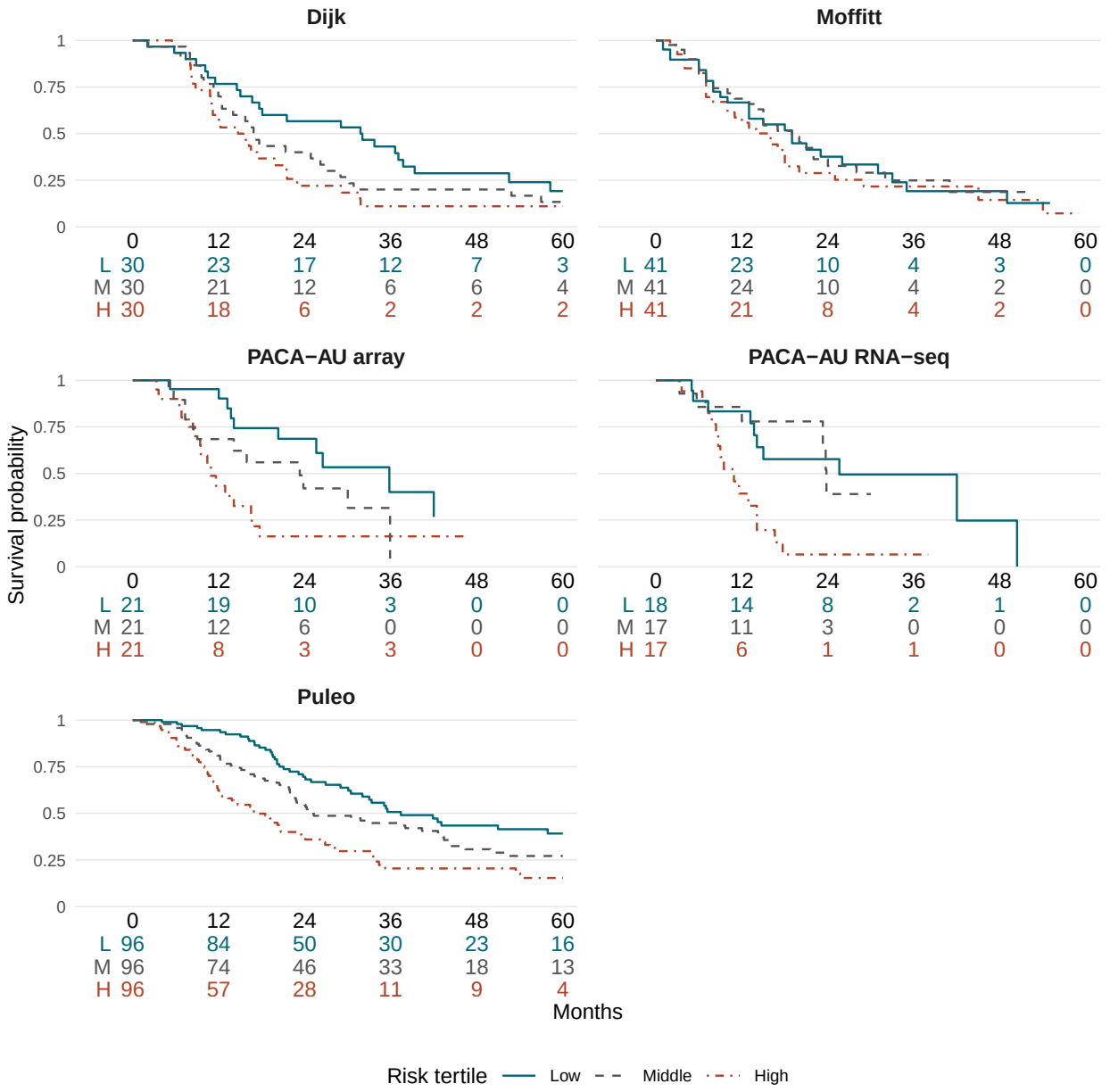


Figure 8: External Kaplan–Meier curves for tertiles of the frozen joint risk score, with numbers at risk.

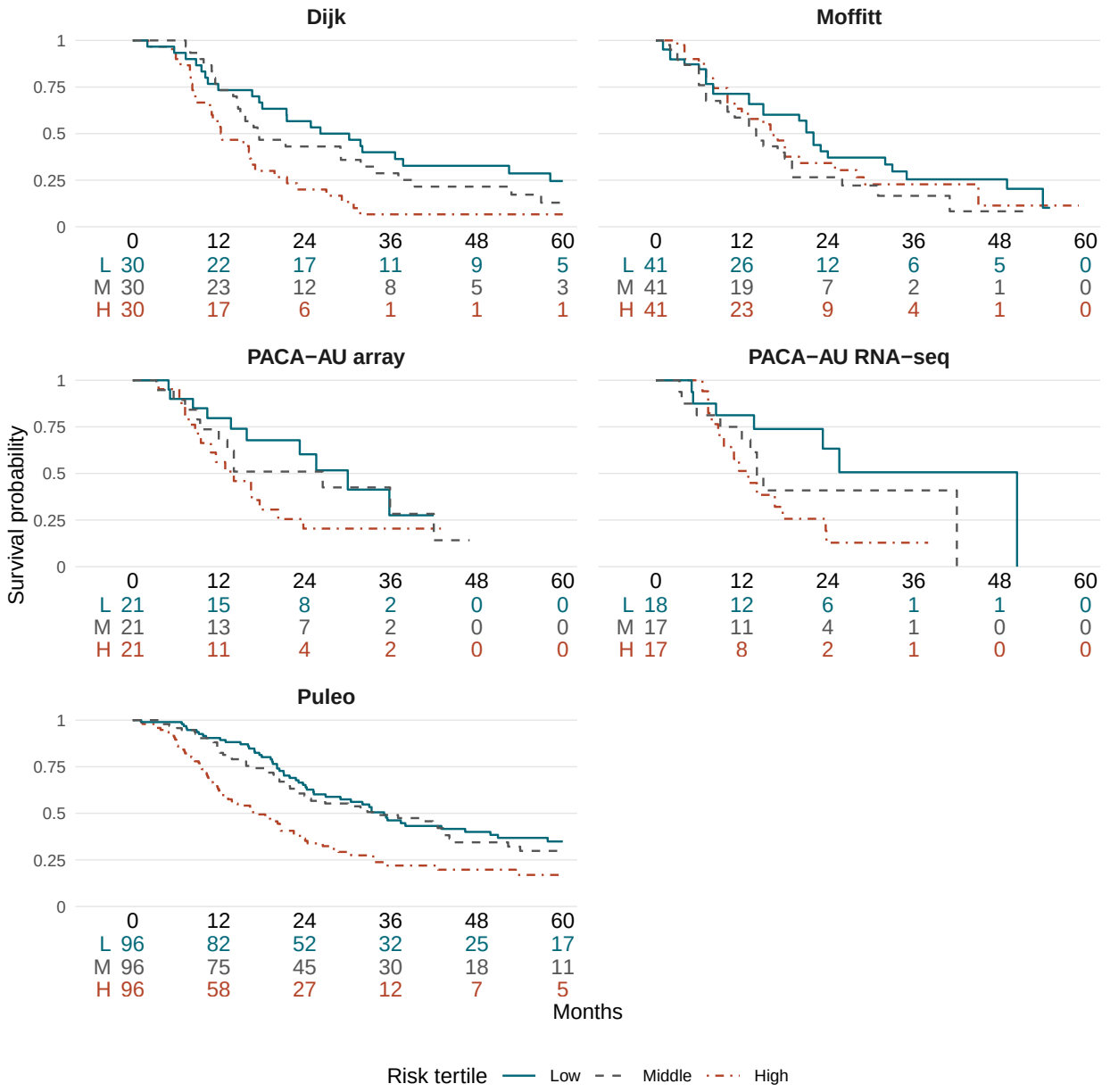


Figure 9: External Kaplan–Meier curves by tertile of the Program 2 score oriented in the adverse training direction, with numbers at risk.

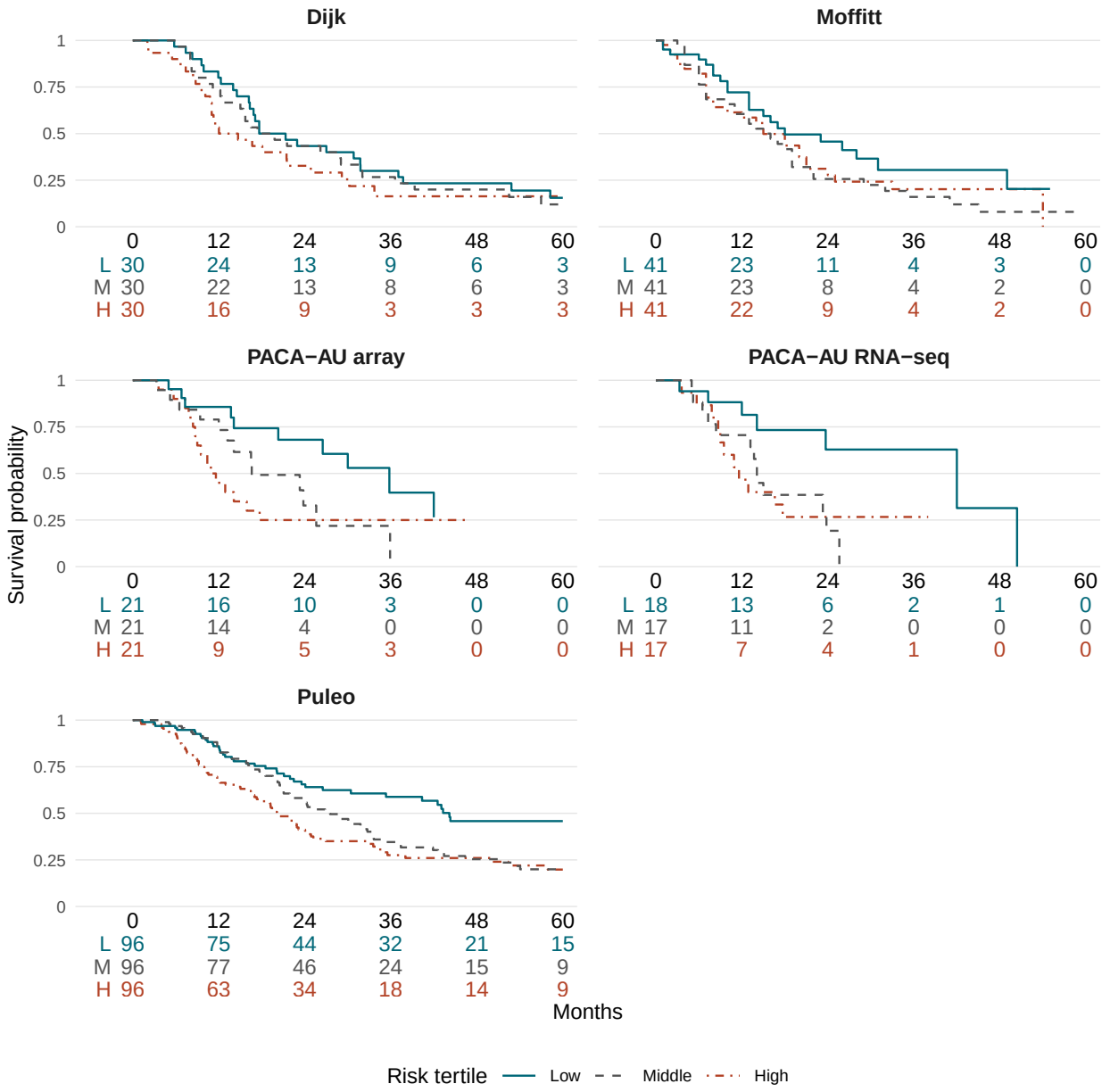


Figure 10: External Kaplan–Meier curves by tertile of the Program 3 score oriented in the adverse training direction, with numbers at risk.

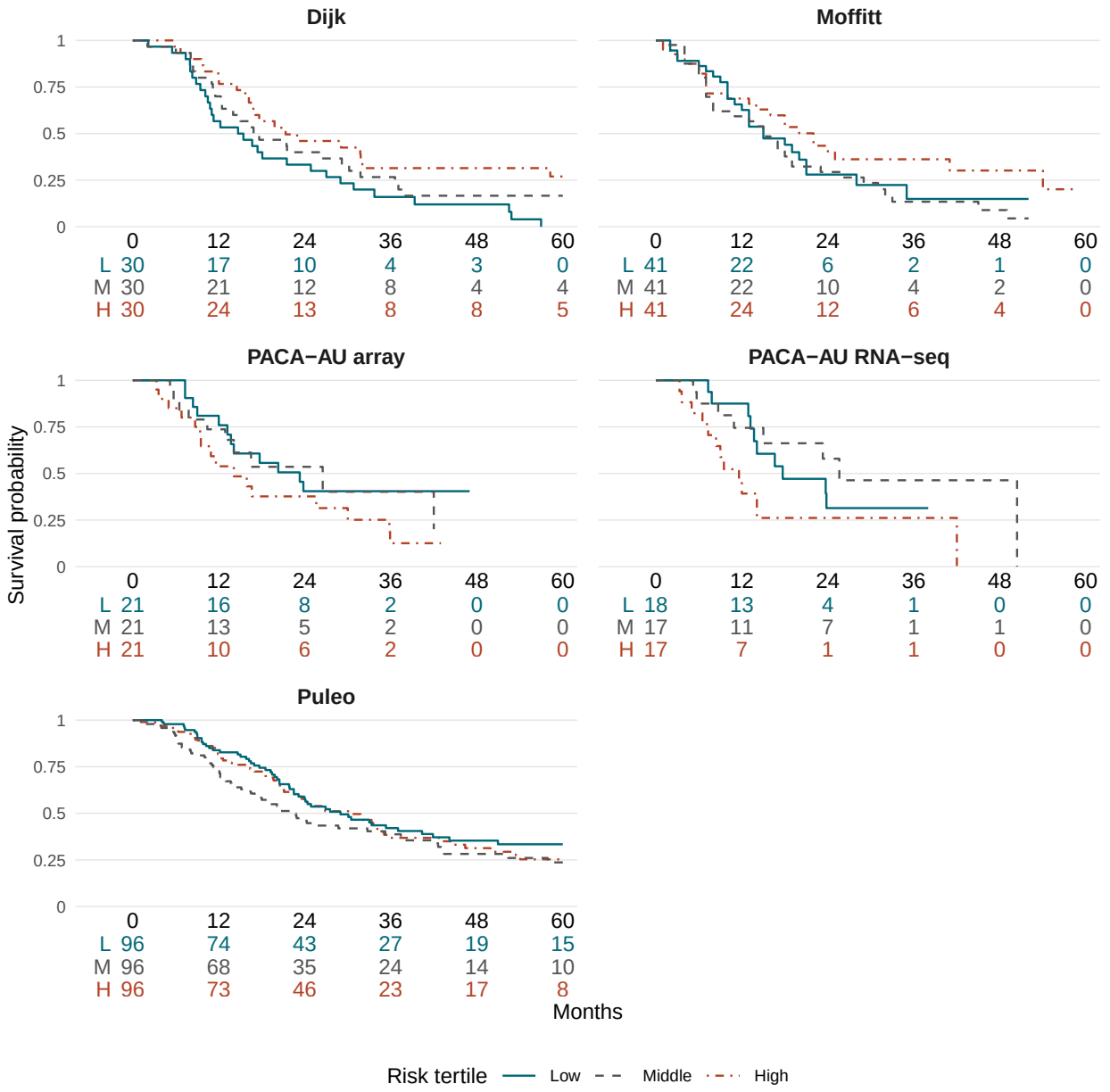


Figure 11: External Kaplan–Meier curves by tertile of the Program 5 score oriented in the adverse training direction, with numbers at risk.

5 Discussion

5.1 Interpretation

The external concordance difference was small and its interval included zero. The estimated loadings were essentially unchanged at the unit survival weight. The joint coefficient fit accounts for posterior loading uncertainty and uses a shrinkage prior. Calibration slopes were closer to one in most series.

In the simulation cells with consistent recovery, the two-step fit already recovered the prognostic

program. The near-threshold simulations isolated a setting where supervision could change discovery. Recovery and concordance improved when survival was strong, while the differences were smaller at realistic survival strength.

5.2 Limitations

Earlier analyses used external concordance to choose the rank carried into this comparison. Each series informed at least one development decision (Section 2.2). The PACA-AU array and RNA-seq records both contribute to the pooled estimate. The donor bootstrap accounts for their shared donors in the interval while retaining both series in the estimand. 2 of 5 supervised training folds stopped at the sweep cap. In simulation the coefficients at $w_S = 1$ are attenuated at the larger gene count. These studies use a common simulation generator.

5.3 Continuing work

Proposed supervision weights include a calibrated power on the Cox partial likelihood, downweighting expression and normalizing the likelihood terms. Each would be assessed for recovery, calibration, false activity and convergence before application to the real data.

The proposed multimodal model shares activities and uses separate loading priors for each data type. A rank-selection rule will be evaluated on training data and simulations. Further comparisons include PurIST (Rashid et al. 2020), DeSurv, elastic-net Cox and a clinical model. An external cohort reserved from all development decisions would support an independent evaluation.

Appendix A. Model derivation

The canonical single-modality procedure uses training expression Y and survival times and indicators. It returns posterior moments of L, F and β, gene precisions and the tracked variational objective.

1. Center and scale the input according to the training preprocessing recipe. Initialize the factorization using a truncated singular value decomposition of the resulting Y.
2. Compute the posterior mean and variance of η. Re-expand the Cox partial log-likelihood about the current mean predictor, using the diagonal curvature to obtain a working normal-means objective.
3. For each program, update β_k using the first and second moments of YF_k. Update L_k from the expression residual. Update the entries of F_k using both likelihood contributions and refresh the projected scores within the loading update. Use empirical Bayes normal-means fits for the prior parameters.
4. Update the gene-specific noise precisions. Record the monitored objective including the weighted Cox term and the Kullback–Leibler terms. Rescale nonzero programs to unit training projection SD and check the relative change in the objective.
5. In each external series, apply the fixed projection loadings and coefficients to the available training genes. Harrell's concordance uses the direction learned from training survival.

The full derivation gives the update formulas, the balance of expression and survival information and the numerical checks. The companion `verify_derivation.R` evaluates individual updates against the expected objective.

Appendix B. Changes since the October 16 version

- **Preprocessing.** Every RNA-seq cohort is logged once. The D4 gene set has 2,218 genes instead of 2,064, 1,985 of them shared.
- **Model.** The proposed model is the survival-weighted fit of Section 1.2. The $\alpha_F = 0$ configuration is a historical reference.
- **Objective.** The Gaussian log-normalizer in the activity KL is counted once per observation. The coefficient KL is exactly zero for a flat coefficient likelihood.
- **Orientation.** Two comparator scripts that chose each cohort's risk direction from its own outcomes now use the training direction. No reported concordance changed.
- **Sign correction.** The post-fit sign flip was removed from both fitters.
- **Simulation calibration.** Calibration slopes in simulation use exact event times.
- **Multimodal model.** Moved to proposed work.

Appendix C. Earlier multimodal results

The October 16 writeup records the previous multimodal simulation and matched expression–methylation comparisons. The proposed extension in this version uses the canonical single-modality objective as its starting point.

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